

GPT-2 smoke run: AUROC comparison

Baselines (4) vs. our feature-ablation variants (12) on Wikipedia held-out + 7 downstream datasets

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Backbone: GPT-2 (124M parameters). **Training set:** 800 MIND-generated Wikipedia records (500 per class, seed=42). **Test split:** 200 records, identical seed across all methods $\Rightarrow$ direct AUROC comparison. **Cells:** AUROC; green-shaded **bold** = column winner.

Methodology (baselines, paper-faithful):

- **SAPLMA** (Azaria & Mitchell 2023): MLP probe 768 $\rightarrow$ 256 $\rightarrow$ 128 $\rightarrow$ 64 $\rightarrow$ 1 on layer-7 last-token hidden. Supervised.
- **HaloScope** (Du et al., NeurIPS 2024): spectral score $\rightarrow$ percentile pseudo-labels $\rightarrow$ non-linear probe 768 $\rightarrow$ 1024 $\rightarrow$ 1. Unsup. pseudo-labels.
- **HalluShift** (Dasgupta, IJCNN 2025): 31-d Wasserstein/cosine on hidden+attention + 11 token-prob features $\rightarrow$ **CombinedNN**. Supervised.
- **EigenScore** / **INSIDE** (Chen et al., ICLR 2024): K=10 sampled responses, K $\times$ K covariance, log-det score. Unsupervised. Downstream subsampled to n=100, K=3.

Type	Method	Wiki	TruthQA	TrivQA	CoQA	TydiQA	HE-QA	HE-Sum	HE-Dial	Avg-MT
B	SAPLMA	0.321	0.531	0.424	0.536	0.518	0.498	0.475	0.576	0.508
B	HaloScope	0.437	0.495	0.545	0.515	0.419	0.089	0.422	0.459	0.421
B	HalluShift	0.443	0.535	0.404	0.583	0.619	0.451	0.561	0.515	0.524
B	EigenScore	0.457	0.505	0.629	0.426	0.455	0.499	0.500	0.500	0.502
V	A: canonical only	0.501	0.515	0.627	0.557	0.563	0.140	0.476	0.364	0.463
V	B: +D_mean	0.405	0.544	0.676	0.413	0.441	0.073	0.459	0.368	0.425
V	C: +V_last	0.394	0.524	0.711	0.484	0.470	0.270	0.542	0.429	0.490
V	D: +H_mean	0.405	0.547	0.679	0.481	0.507	0.187	0.525	0.400	0.475
V	E: +D_mean+V_last+H_mean (MIND+)	0.467	0.533	0.638	0.533	0.565	0.133	0.487	0.372	0.466
V	F: +F1 (Lookback)	0.322	0.494	0.637	0.463	0.391	0.273	0.539	0.532	0.476
V	G: +F5 (Logit-Lens JSD)	0.393	0.500	0.504	0.437	0.423	0.063	0.485	0.395	0.401
V	H: +F7 (Max-Margin)	0.467	0.519	0.705	0.493	0.469	0.052	0.475	0.379	0.442
V	I: +F1+F5+F7 (trio)	0.502	0.537	0.605	0.533	0.576	0.084	0.467	0.358	0.451
V	J: +all 9 F-features	0.475	0.537	0.620	0.538	0.567	0.089	0.492	0.364	0.458
V	K: +MIND+ +F1+F5+F7	0.498	0.516	0.624	0.550	0.550	0.071	0.469	0.355	0.448
V	L: +everything	0.510	0.540	0.569	0.535	0.591	0.041	0.437	0.341	0.436

Table 1: AUROC matrix: 4 SOTA baselines (B) vs. 12 feature-ablation variants (V) on GPT-2. Rows = methods; columns = evaluation datasets. **Bold green** cells mark the per-column winner.

Per-column winners **Notes on interpretation.** GPT-2 (124M) is a deliberately small backbone for this smoke run; signal at this scale is noisy. Headline observations:

Column	Winning method	AUROC
Wikipedia (in-distribution)	[V] L (+everything)	0.5097
TruthfulQA	[V] D (+H_mean)	0.5468
TriviaQA	[V] C (+V_last)	0.7113
CoQA	[B] HalluShift	0.5834
TydiQA-GP	[B] HalluShift	0.6195
HaluEval-QA	[B] EigenScore	0.4994
HaluEval-Summarization	[B] HalluShift	0.5605
HaluEval-Dialogue	[B] SAPLMA	0.5757
Avg multi-task	[B] HalluShift	0.5239

- **HalluShift** wins the average multi-task AUROC (0.524) and 4 of 7 downstream columns.
- Our variants dominate **Wikipedia held-out** (in-distribution) and pick up TruthfulQA + TriviaQA.
- All methods collapse on HaluEval-QA at GPT-2 scale (domain shift; supervised MLPs trained on MIND-Wikipedia are anti-correlated with the right answer).
- For scale-up: variant C (+V_last alone) is the only variant that holds its own against the baselines on transfer (avg-MT 0.490) and is worth re-testing on Qwen-3B.